COCA corporate care

We know ADP believes that people are the heart of every organization

Meet **CoCa** aka CorporateCare, an Employee Sentiment Dashboard designed to help HR teams proactively understand and improve workplace morale.

Your ultimate tool for understanding workplace communication in real time.

In today's fast-paced work environment, employee satisfaction drives productivity.

CoCa empowers HR teams with instant insights to foster a transparent and supportive workplace.

WHY CoCa?

The ADP company challenge highlighted the need for tools that help HR teams proactively improve workplace morale and employee well-being.

Workplace harassment, discrimination, and/or bullying can often fly under the radar as victims hesitate to report.

This issue led us to create CoCa (CorporateCare), designed to offer insights into employee behavior by analyzing workplace communication and flagging toxicity.

The Challenge

Manually sorting through these emails,text messages and feedback forms, identifying urgent concerns, and recognizing underlying sentiment trends is **time-consuming** and **reactive**.

CoCa is the solution, a dashboard with an AI powered-model tracking overall sentiment trends, **flagging discrimination**, and breaking down employee engagement.

ACTION PLAN

Armed with this data, HR can now:

- **Address workplace discrimination** concerns by revisiting policies.

- **Investigate** flagged messages and mediate conflicts before they escalate.

- **Encourage** more manager feedback to improve communication culture.

The BERT Model

We used a sentiment detection model from Hugging Face and trained it with data to detect ***discrimination*** (different types of workplace discrimination)

Data Preparation:

- Cleaned employee messages (lowercase, removed special characters, stop words, and duplicates).
- Converted target variable (is_discrimination) to none values (0 for "no", 1 for "yes").
- Mapped discrimination types to numerical values (sexist: 0, racist: 1, none: 2).

Model Architecture:

- Used a multi-output BERT model:
  - Output 1: Predicts if a message is discriminatory (yes/no).
  - Output 2: Predicts the type of discrimination (sexist, racist, or none).

Your paragraph text

TRAINING:

Your paragraph text

- Split data into training and validation sets.
- Trained the model and used cross-entropy loss with the AdamW optimizer to avoid overfitting.

EVALUATION

```
Classification Report for is_discriminatory:
              precision    recall  f1-score   support

           0       0.88      0.64      0.74        11
           1       0.67      0.89      0.76         9

    accuracy                           0.75        20
   macro avg       0.77      0.76      0.75        20
weighted avg       0.78      0.75      0.75        20

Classification Report for discrimination_type:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00         4
           1       0.50      0.80      0.62         5
           2       0.88      0.64      0.74        11

    accuracy                           0.55        20
   macro avg       0.46      0.48      0.45        20
weighted avg       0.61      0.55      0.56        20
```

DASHBOARD DEMO!!

FUTURE PLANS

1. Data Augmentation and Preprocessing
   - Collect more labeled data, especially for under-represented types of discrimination. Create more diverse synthetic data
   - Focus on subtle or implicit discrimination examples.

2. Model Architecture and Training
   - Combine predictions from different models to improve accuracy.
   - Use grid search or Bayesian optimization to find the best settings.
   - Use k-fold cross-validation to improve reliability and reduce overfitting.
   - Monitor performance and fairness over time.

3. Add More Classifications:
   - Add more discrimination types to identify
   - Classify messages based on severity type

Instead of waiting for problems to surface, HR can listen, learn, and act—before it's too late.

Because a thriving company starts with a thriving workforce with a thriving CoCa!!

Thank You!